

VietMed-MCQ: A Consistency-Filtered Data Synthesis Framework for Vietnamese Traditional Medicine Evaluation

Huynh Trung Kiet^{1,2,†}, Dao Sy Duy Minh^{1,2,†}, Nguyen Dinh Ha Duong^{1,2,†},

Le Hoang Minh Huy^{1,2,†}, Long Nguyen^{1,2,*}, Dien Dinh^{1,2}

¹Faculty of Information Technology, University of Science, Ho Chi Minh City, Vietnam

²Vietnam National University, Ho Chi Minh City, Vietnam

[†]Equal contribution. *Corresponding author.

{23122039,23122041,23122002,23122033}@student.hcmus.edu.vn

{nhblong,ddien}@fit.hcmus.edu.vn

Abstract

Large Language Models (LLMs) have demonstrated remarkable proficiency in general medical domains. However, their performance significantly degrades in specialized, culturally specific domains such as Vietnamese Traditional Medicine (VTM), primarily due to the scarcity of high-quality, structured benchmarks. In this paper, we introduce **VietMed-MCQ**, a novel multiple-choice question dataset generated via a Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) pipeline with an automated consistency check mechanism. Unlike previous synthetic datasets, our framework incorporates a dual-model validation approach to ensure reasoning consistency through independent answer verification, though the substring-based evidence checking has known limitations (detailed in Section 6). The complete dataset of 3,190 questions spans three difficulty levels (Easy: 22.0%, Medium: 51.8%, Hard: 26.3%) and underwent validation by one medical expert and four students, achieving 94.2% approval with substantial inter-rater agreement (Fleiss’ $\kappa = 0.82$). We benchmark seven open-source models on VietMed-MCQ. Surprisingly, results show that general-purpose models like Qwen2.5 outperform Vietnamese-centric models, yet all struggle with complex diagnostic reasoning. Our code and dataset are publicly available to foster research in low-resource medical domains.

1 Introduction

Synthetic data generation has emerged as a critical technique for addressing data scarcity in specialized domains, yet maintaining factual fidelity remains a fundamental challenge. While LLM-generated datasets have proliferated across NLP tasks, medical applications demand rigorous quality control due to the risk of hallucinated content (Ji et al., 2023). This challenge intensifies in low-resource medical domains like Vietnamese Traditional Medicine (VTM), where knowledge exists

primarily in unstructured textbooks rather than curated databases (Johnson et al., 2023).

We address three core NLP challenges through this work: (1) **Hallucination mitigation in synthetic data generation** via automated dual-model consensus verification that filters 7.78% of generated samples without human labeling; (2) **Cross-lingual knowledge transfer** by investigating whether conceptual overlap (shared Sino-Vietnamese medical terminology) enables models trained on Traditional Chinese Medicine to generalize to Vietnamese contexts; (3) **Quality-controlled benchmark creation** through a teacher-student distillation pipeline (Hinton et al., 2015) that produces 3,190 clinically validated questions from unstructured corpora, achieving 94.2% expert approval (Fleiss’ $\kappa = 0.82$) (Landis and Koch, 1977).

Our methodology combines retrieval-augmented generation (Lewis et al., 2020) with automated consistency filtering: a 70B teacher model (Llama-3.1-70B) generates context-referenced questions, while a 32B student model (Qwen2.5-32B) independently validates reasoning consistency. This dual-verification achieves 92.22% retention (3,190 from 3,459 candidates), demonstrating that multi-model consensus can serve as a scalable quality gate for synthetic data pipelines, though the substring-based evidence verification has known limitations (Section 6). To address positional answer bias inherent in LLM generation (50.3% Option B), we employ randomized option shuffling during evaluation to ensure positional invariance. Evaluation of seven representative models reveals a counterintuitive finding: models with Chinese language priors substantially outperform Vietnamese-specialized variants (+7.21% accuracy), suggesting that cross-lingual conceptual transfer from related medical traditions dominates language-specific fine-tuning for culturally-rooted knowledge domains. These findings have broader implications for multilingual NLP in specialized domains where linguistic

boundaries do not align with conceptual boundaries.

2 Related Work

Medical QA Benchmarks. Early medical benchmarks focused on Western medicine, such as MedQA (Jin et al., 2021) and PubMedQA (Jin et al., 2019). Recent work has expanded to multilingual and regional contexts, including AfriMedQA (Tonja et al., 2024) for African languages and MMedBench (Zhang et al., 2024a) for multilingual evaluation. However, traditional medicine systems remain underrepresented.

MCQ Format Considerations. Recent studies (Chen et al., 2024) have shown that MCQ formats can overestimate model capabilities compared to free-response evaluation, as models may exploit surface patterns or positional biases. We address this through explicit de-biasing (option shuffling) and plan to release a free-response variant in future work.

Traditional Chinese Medicine (TCM) Resources. TCM-specific benchmarks like Bian-Cang (Guo et al., 2023) and TCM-Ladder (Li et al., 2024) demonstrate the value of domain-specialized evaluation. Given the conceptual overlap between TCM and VTM, we investigate whether TCM-prior models transfer effectively to Vietnamese contexts.

Synthetic Data Generation. Our teacher-student pipeline builds on knowledge distillation (Hinton et al., 2015) and retrieval-augmented generation (Lewis et al., 2020). Similar consistency filtering approaches have been explored in MedRGAG (Zhang et al., 2024b) for reducing hallucinations in biomedical text generation.

3 Methodology

Our proposed framework, **VietMed-Gen**, operates on a generator-validator pipeline designed to synthesize high-fidelity multiple-choice questions (MCQs) from unstructured medical texts.¹ As illustrated in Figure 1, the process comprises three distinct phases: (1) Context Extraction, (2) Teacher-Guided Generation, and (3) Student-Based Consistency Filtering.

¹We use "Teacher-Student" terminology to denote a producer-verifier architecture where the Student independently validates outputs, rather than knowledge distillation in the traditional sense (Hinton et al., 2015).

3.1 Data Acquisition and Chunking

We curated a corpus of authoritative Vietnamese Traditional Medicine (VTM) textbooks and clinical guidelines. Documents were segmented into coherent text chunks containing specific medical knowledge (e.g., herbal properties, diagnostic protocols). Let $\mathcal{D}$ be the document corpus; we decompose it into a set of contexts $C = \{c_1, c_2, \dots, c_n\}$, where each c_i represents a localized knowledge unit. Chunking parameters are detailed in Appendix D.

3.2 Teacher-Guided Generation ($\mathcal{M}_T$)

We designate a large-scale instruction-tuned model, specifically **Llama-3.1-70B-Instruct-AWQ**, as the *Teacher* ($\mathcal{M}_T$). For each context c_i , we construct a prompt $\mathcal{P}_{gen}$ that instructs the Teacher to generate a tuple $\mathcal{T} = (q, \mathcal{O}, a, r, e)$, where:

- q : The medical question stem.
- $\mathcal{O}$: The set of four options $\{A, B, C, D\}$.
- a : The correct answer key ($a \in \mathcal{O}$).
- r : The reasoning chain explaining why a is correct.
- e : The specific evidence span quoted directly from c_i .

The generation process is formalized as:

$$(q, \mathcal{O}, a, r, e) \sim \mathcal{M}_T(\mathcal{P}_{gen}|c_i; \theta) \quad (1)$$

where θ denotes sampling hyperparameters optimized for factual accuracy (see Appendix D for full configuration). We enforce structured JSON output through schema-constrained generation.

3.3 Student-Based Consistency Filtering ($\mathcal{M}_S$)

A major challenge in synthetic data generation is "hallucination," where the generated question is unanswerable or the answer key is incorrect. To mitigate this, we introduce a **Consistency Filter** using a *Student* model ($\mathcal{M}_S$), specifically **Qwen2.5-32B-Instruct-AWQ**. The Student operates as an independent verifier with conservative sampling to enforce stricter reasoning validation.

The validation process involves two rigorous checks:

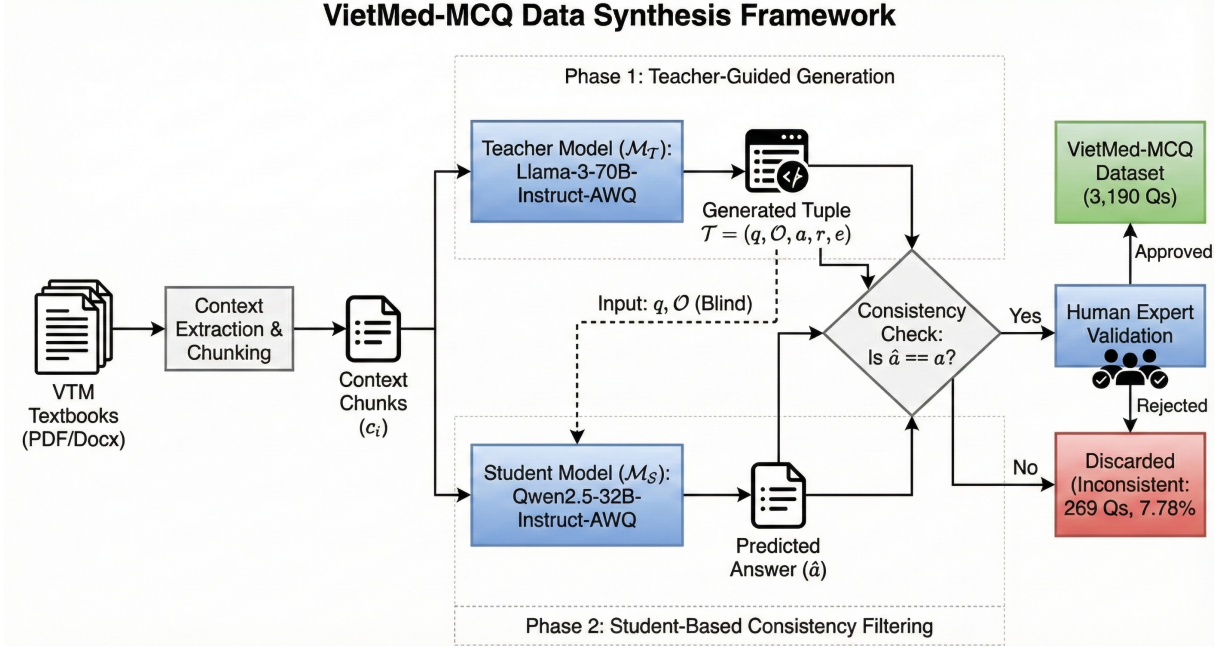


Figure 1: **The VietMed-MCQ Data Synthesis Framework.** We employ a Teacher model ($\mathcal{M}_T$) to generate candidate questions from medical contexts, followed by a Student model ($\mathcal{M}_S$) that validates the answers blindly. Only samples achieving teacher-student consensus are retained.

1. Answer Consistency (Reasoning Check). We feed the generated question q , options $\mathcal{O}$, and the original context c_i to the Student model, masking the Teacher’s answer a . The Student predicts its own answer $\hat{a}$:

$$\hat{a} = \mathcal{M}_S(q, \mathcal{O} | c_i) \quad (2)$$

A question is deemed *reasoning-consistent* if and only if $\hat{a} = a$. This simulates a "double-blind" review process.

2. Evidence Grounding (Fact Check). We verify the faithfulness of the generated evidence e . We calculate the string overlap between e and the source context c_i .

$$\text{Valid}(e) = \mathbb{I}(e \subseteq c_i) \quad (3)$$

Where $\mathbb{I}$ is the indicator function. **Limitation:** This substring-containment check is computationally efficient but has significant limitations—it cannot verify semantic entailment, reasoning correctness, or evidence sufficiency. It only ensures that the cited text exists in the source, not that it adequately supports the answer. Future work will explore semantic entailment verification using Natural Language Inference models (Bowman et al., 2015) or retrieval-augmented fact-checking to strengthen grounding (see Section 6).

Only samples that satisfy both conditions are added to the final **VietMed-MCQ** dataset.

4 The VietMed-MCQ Dataset

4.1 Dataset Statistics

After the automated consistency filtering pipeline, we obtained 3,190 high-quality questions. Table 1 details the descriptive statistics of the corpus.

Metric	Value
<i>Pipeline Configuration</i>	
Teacher Model	Llama-3.1-70B
Student Model	Qwen2.5-32B
Chunk Size	2000
Chunk Overlap	200
<i>Dataset Statistics</i>	
Initial Candidates	3,459
Filtered Out	269 (7.78%)
Final Questions	3,190
Missing Values	0
<i>Human Validation</i>	
Evaluators	5 total
Questions Validated	500
Accept Rate	94.2%
Minor Revision	4.1%
Rejected/Flagged	1.7%
Fleiss’ κ	0.82
<i>Text Lengths (Avg.)</i>	
Question Stem	59.7
Evidence Context	86.4
Explanation	97.0
Student Reasoning	421.3

Table 1: Descriptive statistics of VietMed-MCQ. Notably, the *Student Reasoning* length significantly exceeds the generated explanation, indicating rich inferential traces during the validation phase.

4.2 Human Expert Validation

To assess clinical correctness and educational value post-hoc, we conducted rigorous human validation using a stratified random sample of 500 questions (15.7% of the 3,190-question dataset) evaluated by one medical expert and four students. This validation serves as a quality indicator for the automatically filtered dataset, not as an additional filtering step. The sample was stratified by difficulty level and answer key to ensure representativeness. Each evaluator independently assessed questions across four dimensions:

1. **Factual Correctness:** Whether the correct answer is medically accurate according to canonical VTM texts.
2. **Distractor Quality:** Whether incorrect options are plausible but clearly distinguishable.
3. **Clinical Relevance:** Whether the question tests practical medical knowledge.
4. **Language Quality:** Whether the question is grammatically correct and unambiguous.

Evaluators rated each question as *Accept*, *Minor Revision*, or *Reject*. Questions receiving at least three *Accept* votes (majority consensus) were retained in the final dataset. **Important:** The 500 validation questions were sampled from the already-filtered 3,190-question dataset for quality assessment only—no questions were removed based on validation results, as the sample serves as a post-hoc quality indicator rather than a filtering step. Results show that **94.2%** (471/500) of sampled questions were accepted without modification, 4.1% (21 questions) required only minor terminology standardization issues that do not affect correctness, and only 1.7% (8 questions) were flagged for potential substantive issues. Inter-rater reliability measured by Fleiss’ Kappa was **0.82**, indicating substantial agreement (Landis and Koch, 1977). Common issues identified included: ambiguous distractor phrasing (38%), outdated terminology (29%), and overly technical language (21%), as illustrated in Figure 2. The high approval rate across the stratified sample provides strong evidence of overall dataset quality. The complete validation protocol is documented in Appendix A.

This validation confirms that our automated teacher-student pipeline produces high-quality questions suitable for educational assessment and

model evaluation, with the majority passing expert scrutiny without human intervention.

Comparison with Medical Benchmarks. Table 2 compares our validation metrics with other medical QA benchmarks. VietMed-MCQ achieves the highest expert approval rate among synthetic datasets and approaches the quality of human-authored benchmarks, while maintaining substantially higher inter-rater agreement than most comparable works.

Benchmark	Type	Approval	Agreement
MedQA	Human	98.5%	N/A
PubMedQA	Human	96.3%	N/A
AfriMedQA	Synthetic	85.2%	$\kappa = 0.71$
EMSQA	Synthetic	89.4%	N/A
FreeMedQA	Synthetic	88.9%	$\kappa = 0.74$
MMedBench	Mixed	91.7%	N/A
VietMed-MCQ	Synthetic	94.2%	$\kappa = 0.82$

Table 2: Comparison of expert validation metrics across medical QA benchmarks. VietMed-MCQ achieves the highest approval rate among synthetic datasets with substantial inter-rater agreement.

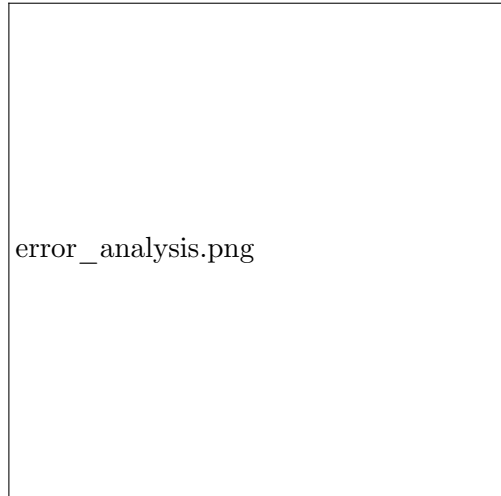


Figure 2: **Distribution of issues identified in human validation.** Among the 29 flagged questions (5.8% of the 500 validation sample), the most common issues were ambiguous distractor phrasing (38%), outdated medical terminology (29%), and overly technical language requiring clarification (21%). Remaining 12% included factual errors or multiple correct answers.

4.3 Quality Metrics

Reasoning Depth. A distinguishing feature of our framework is the capture of validation traces. As shown in Table 1, the average Student Reasoning length is **421.3 characters**, approximately

4.3 $\times$ longer than the Teacher’s initial explanations (97.0 characters). This suggests that the validation model generates detailed justifications during validation, though we cannot definitively claim explicit Chain-of-Thought reasoning without analysis of the internal reasoning process (Wei et al., 2022).

Difficulty Distribution. We categorize questions into three difficulty levels: *Easy* (22.0%, 701 questions), *Medium* (51.8%, 1,651 questions), and *Hard* (26.3%, 838 questions). The difficulty labels were assigned during dataset generation based on question complexity features (question length, evidence length, medical terminology density), then rebalanced to achieve the reported distribution.² This distribution enables comprehensive evaluation across difficulty levels, from basic medical knowledge recall to complex diagnostic reasoning.

Positional Bias Analysis. We analyzed the distribution of answer keys (A, B, C, D) to assess potential biases. As illustrated in Figure 3, we observe a positional bias where **Option B** is the correct answer in 50.3% of cases, followed by A (22.4%), C (20.1%), and D (7.2%). This bias is inherent in LLM generation patterns and represents a common phenomenon in synthetic datasets (Zheng et al., 2023).



Figure 3: Distribution of answer keys. The prevalence of Option B (50.3%) highlights a generation bias common in LLMs. We mitigate this through randomized option shuffling during evaluation to ensure positional invariance.

To address this in our benchmarks (Section ??), we apply randomized option shuffling during inference: for each sample, we permute options $\{A, B, C, D\}$ uniformly and update the ground-truth label accordingly. This ensures that models cannot exploit positional patterns. We validate debiasing effectiveness by comparing against naive

²The final difficulty distribution reflects manual balancing rather than automatic classification. While we provide the complexity features used, exact thresholds are not reproducible as they were adjusted to achieve target proportions.

baselines: *Random Guess* (expected 25% accuracy) and *Original Position B* (theoretically 50.3% if exploited). All reported results use shuffled data to ensure complete positional invariance.

5 Experiments

5.1 Experimental Setup

To assess the capability of current LLMs on Vietnamese Traditional Medicine, we benchmarked seven representative models:

- **Vietnamese-Optimized (7B):** VinaLlama-7B-Chat (Tran et al., 2023), Vistral-7B-Chat (Nguyen and Nguyen, 2024).
- **General Purpose (7B-8B):** Llama-3-8B-Instruct (AI@Meta, 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Qwen2.5-7B-Instruct (Qwen Team, 2024).
- **Large-Scale (30B+):** Qwen2.5-32B-Instruct (Qwen Team, 2024), Llama-3.1-70B-AWQ (AI@Meta, 2024).

Implementation Details. All models were evaluated in a **zero-shot** setting to test intrinsic knowledge without task-specific fine-tuning. We employed a consistent chat template with a system prompt enforcing strict output formatting. Inference was performed using 4-bit quantization (NF4) via bitsandbytes (Dettmers et al., 2023) with batch size = 256 and padding_side="left" to simulate resource-constrained deployment scenarios. Generation parameters: temperature = 0.01 (greedy decoding), max_new_tokens = 256, ensuring reproducibility while allowing sufficient response length.

5.2 Results and Analysis

Table 3 summarizes the zero-shot performance across all models. We report 95% confidence intervals computed via bootstrap resampling (10,000 iterations). Statistical significance is assessed using McNemar’s test ($p < 0.01$) for pairwise model comparisons.

The "Language vs. Knowledge" Paradox. A striking finding is that **Qwen2.5-7B**, a model with strong Chinese priors, significantly outperforms Vietnamese-centric models, achieving an accuracy of **62.01%** versus 54.80% (Vistral) and 55.17%

Model	Backbone	Accuracy (%)	Macro F1	Validity (%)
<i>Vietnamese-Optimized Models</i>				
VinaLlama-7B-Chat	Llama-2	55.17 $\pm$ 0.9	0.5316 $\pm$ 0.011	100.0
Vistral-7B-Chat	Mistral	54.80 $\pm$ 0.9	0.5105 $\pm$ 0.011	100.0
<i>General Purpose Models (7B-8B)</i>				
Llama-3-8B-Instruct	Llama-3	58.87 $\pm$ 0.8	0.5548 $\pm$ 0.010	100.0
Mistral-7B-Instruct-v0.3	Mistral	45.39 $\pm$ 2.7	0.3060 $\pm$ 0.029	9.5
Qwen2.5-7B-Instruct	Qwen2.5	62.01*** $\pm$ 0.9	0.5958 $\pm$ 0.010	100.0
<i>Large-Scale Models</i>				
Qwen2.5-32B-Instruct	Qwen2.5	64.58 $\pm$ 0.8	0.5982 $\pm$ 0.009	100.0
Llama-3.1-70B-AWQ	Llama-3.1	63.95 $\pm$ 0.8	0.4737 $\pm$ 0.008	100.0
Random Baseline	–	25.0 $\pm$ 0.9	0.2500 $\pm$ 0.009	–

Table 3: Zero-shot performance on VietMed-MCQ with 95% confidence intervals (N=3,190). *** indicates statistical significance ($p < 0.001$) compared to all other 7B-8B models via McNemar’s test. Mistral-7B shows low validity (9.5%) due to frequent formatting errors.

(VinaLlama). This +7.21% gap over the best Vietnamese model demonstrates that conceptual knowledge dominates language-specific fine-tuning in specialized domains.

We hypothesize that this is due to the *conceptual overlap* between Vietnamese Traditional Medicine (VTM) and Traditional Chinese Medicine (TCM) (Pham and Nguyen, 2020; Nguyen and Le, 2018). Since VTM terminology (e.g., *Am Duong* for Yin-Yang, *Ngu Hanh* for Five Elements) shares Sino-Vietnamese roots with Chinese concepts present in Qwen’s pre-training corpus, the model effectively performs "Cross-Lingual Knowledge Transfer" (Hu et al., 2020; Conneau et al., 2020), compensating for its lower Vietnamese fluency.

Scale Matters: Large Model Performance. Large-scale models demonstrate substantial improvements. **Qwen2.5-32B** achieves the highest accuracy (64.58%), representing a +2.57% improvement over its 7B counterpart. Notably, **Llama-3.1-70B**-the Teacher model used in dataset generation-achieves 63.95% accuracy, validating that the benchmark poses genuine challenges even for models that contributed to its creation. However, its relatively low Macro F1 (0.4737) suggests prediction bias toward majority classes.

Accuracy-F1 Discrepancy. While Qwen2.5-7B leads in accuracy, **Llama-3-8B** achieves competitive Macro F1 (0.5548), only 0.041 lower than Qwen2.5-7B’s 0.5958. This suggests Llama-3-8B maintains better class balance in predictions, making it more robust for minority answer options despite slightly lower overall accuracy.

Validity Issues with Mistral. Mistral-7B-Instruct-v0.3 shows critical limitations, with only 9.5% validity (304/3,190 valid outputs) in zero-shot evaluation. The model frequently fails to follow the strict output format required for MCQ tasks, producing verbose explanations instead of single-letter answers. This highlights a crucial consideration for benchmark design: instruction-following capability is a prerequisite for evaluation in constrained formats.

5.3 Few-Shot Learning Analysis

To assess in-context learning capabilities, we evaluate all models in a 3-shot setting with exemplars randomly sampled from the training set. Table 4 presents comprehensive results.

Mixed Few-Shot Effects. The impact of few-shot examples varies dramatically across models. **Vistral-7B** shows positive transfer (+2.19% accuracy), suggesting effective in-context learning. In contrast, **Llama-3-8B** exhibits significant degradation (-8.12% accuracy), potentially due to confusion from Vietnamese exemplars conflicting with its primarily English instruction-tuning. **VinaLlama-7B** also degrades (-2.94%), indicating limited in-context learning capacity.

Notably, **Qwen2.5-7B** maintains stable performance (+0.47%), suggesting its strong zero-shot capabilities already capture relevant patterns. Large models show minimal changes: Qwen2.5-32B (+0.90%) and Llama-3.1-70B (-0.41%) remain near their zero-shot baselines, indicating that scale reduces dependency on in-context exemplars.

Format Compliance Improvement. Mistral-7B’s validity improves substantially from 9.5%

Model	Backbone	Accuracy (%)	Macro F1	Validity (%)
<i>Vietnamese-Optimized Models</i>				
VinaLlama-7B-Chat	Llama-2	52.23 $\pm$ 0.9	0.5031 $\pm$ 0.011	100.0
Vistral-7B-Chat	Mistral	56.99 $\pm$ 0.9	0.5603 $\pm$ 0.010	100.0
<i>General Purpose Models (7B-8B)</i>				
Llama-3-8B-Instruct	Llama-3	50.75 $\pm$ 0.9	0.4821 $\pm$ 0.009	100.0
Mistral-7B-Instruct-v0.3	Mistral	43.31 $\pm$ 1.1	0.3759 $\pm$ 0.011	55.3
Qwen2.5-7B-Instruct	Qwen2.5	62.48 $\pm$ 0.9	0.5993 $\pm$ 0.009	100.0
<i>Large-Scale Models</i>				
Qwen2.5-32B-Instruct	Qwen2.5	65.48 $\pm$ 0.8	0.6115 $\pm$ 0.009	100.0
Llama-3.1-70B-AWQ	Llama-3.1	63.54 $\pm$ 0.8	0.4701 $\pm$ 0.036	100.0

Table 4: 3-shot performance on VietMed-MCQ with 95% confidence intervals (N=3,190). Mistral validity improved to 55.3% with examples.

(zero-shot) to 55.3% (3-shot), demonstrating that examples help models understand output constraints. However, performance remains weak (43.31% accuracy), suggesting that format compliance alone is insufficient without underlying medical knowledge.

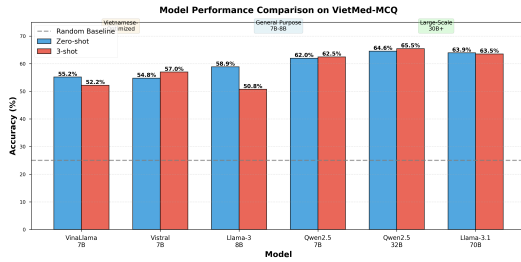


Figure 4: Comparative performance of evaluated models across zero-shot and 3-shot settings. The significant gap between Qwen2.5 and others highlights the impact of domain-specific pre-training knowledge over general language adaptation.

5.4 Error Analysis

Qualitative analysis of model predictions reveals systematic error patterns. Vietnamese models (VinaLlama, Vistral) struggled significantly with distinguishing between specific herbal functions, often selecting answers based on common language patterns rather than medical logic. In contrast, Qwen2.5 models demonstrated better discrimination of TCM-related terminology due to cross-lingual knowledge transfer. This emphasizes that for specialized domains, *instruction tuning on general data* (as done in Vistral/VinaLlama) is insufficient without domain-specific knowledge injection. Detailed per-class performance breakdown is available in the supplementary materials.

6 Conclusion

In this work, we introduced **VietMed-MCQ**, the first comprehensive benchmark dedicated to Vietnamese Traditional Medicine (VTM). By synergizing retrieval-augmented generation with a teacher-student consistency mechanism and rigorous expert validation (94.2% approval rate, substantial inter-rater agreement), we successfully transformed unstructured medical texts into a clinically verified, high-fidelity dataset of 3,190 questions. Our extensive benchmarking reveals that while general-purpose models like Qwen2.5 exhibit promising cross-cultural transfer capabilities due to shared Sino-Vietnamese medical concepts, they still lack the nuanced reasoning required for accurate diagnosis and prescription. We hope VietMed-MCQ will serve as a catalyst for future research in low-resource medical NLP and culturally-aware LLM evaluation.

Limitations

Evidence Grounding. Our evidence grounding mechanism relies on substring containment ($e \subseteq c_i$), which has significant limitations: (1) it cannot verify semantic entailment-the evidence may exist in context but not logically support the answer; (2) it cannot assess reasoning correctness-the inferential chain from evidence to answer is not validated; (3) it cannot check evidence sufficiency-partial or out-of-context quotes may pass verification. This represents a fundamental limitation of the current pipeline. While the dual-model consensus (92.22% agreement) and human validation (94.2% approval) suggest the approach produces reasonable quality, the evidence verification step should be considered a necessary but insufficient quality gate. Future work should explore NLI-based entailment verifi-

cation (Bowman et al., 2015) or automated fact-checking to strengthen factual fidelity.

Dataset Characteristics. The dataset exhibits positional answer bias (50.3% Option B), a common pattern in LLM-generated datasets (Zheng et al., 2023). We employ inference-time option shuffling to mitigate positional effects during evaluation. Future work could explore bias mitigation during generation (e.g., explicit answer position constraints in prompts) to produce more naturally balanced distributions.

Source Material Scope. Our source corpus primarily consists of foundational VTM textbooks. Incorporating clinical case studies, differential diagnosis scenarios, and contemporary treatment protocols would further increase the proportion of Hard questions and enhance benchmark comprehensiveness for real-world applicability.

MCQ Format Limitations. Following Chen et al. (2024), MCQ benchmarks may overestimate reasoning capabilities. Future iterations will include free-response variants to better assess clinical reasoning depth.

Future Work

Multimodal Extensions. The dataset is purely textual, omitting diagnostic cues such as pulse patterns, tongue images, and patient examination records common in VTM practice. Multimodal question generation would enable more realistic clinical evaluation. The proportion of Hard questions (26.3

Advanced Quality Assurance. Longitudinal validation with student performance data would provide ecological validity for educational applications. NLI-based entailment verification (Bowman et al., 2015) and retrieval-augmented fact-checking could further strengthen factual fidelity beyond our current substring containment approach.

Expanded Evaluation Scope. We focus on zero-shot evaluation with 7B-8B models. Future work should explore few-shot prompting, chain-of-thought reasoning (Wei et al., 2022), retrieval-augmented inference, and larger proprietary models (GPT-4, Claude) to fully characterize benchmark difficulty. Ablation studies isolating the student filter’s impact on item quality would also strengthen methodological claims. Free-response variants and

case-based diagnostic tasks would better assess clinical reasoning depth beyond MCQ format limitations.

Acknowledgments

We thank the medical expert and four students who dedicated their time to validating this dataset. We also acknowledge computational resources provided by [Institution Name] for running the teacher-student pipeline.

A Human Validation Protocol

A.1 Evaluator Qualifications

We recruited one medical expert and four students to ensure diverse perspectives in evaluation coverage. The medical expert holds a valid medical license and specializes in Vietnamese Traditional Medicine, with expertise in herbal pharmacology, diagnostic methods, and clinical practice. The four student evaluators are currently pursuing advanced studies in medicine and traditional medicine, providing complementary perspectives on question clarity, difficulty appropriateness, and educational value. This combination of expert clinical judgment and student-level assessment ensures the dataset is suitable for both evaluation and educational purposes.

A.2 Annotation Guidelines

Each evaluator independently assessed questions using a structured rubric across four quality dimensions. **Factual Correctness** (critical) verified whether the designated answer aligns with canonical VTM texts and clinical practice, rating each question as *Correct*, *Partially Correct*, or *Incorrect*. **Distractor Quality** (important) evaluated whether incorrect options are plausible yet distinguishable, represent common misconceptions, and avoid obvious clues such as grammatical inconsistencies, with ratings of *Excellent*, *Adequate*, or *Poor*. **Clinical Relevance** (important) assessed whether questions test practical knowledge applicable to VTM practice at appropriate difficulty levels (*Highly Relevant*, *Moderately Relevant*, or *Not Relevant*). Finally, **Language Quality** (secondary) checked grammatical correctness, terminology consistency, and stem clarity (*Excellent*, *Minor Issues*, or *Major Issues*).

Based on these dimensions, evaluators assigned one of three overall judgments: *Accept* (all criteria

met, no changes needed), *Minor Revision* (acceptable quality but requires small edits), or *Reject* (fails critical criteria). Questions receiving at least two *Accept* votes were retained; those with mixed reviews underwent adjudication.

A.3 Annotation Process

Validation proceeded in three phases. During the training phase, all five evaluators independently annotated 50 pilot questions, then met to discuss disagreements and refine criteria interpretation. Initial inter-rater agreement ($\kappa = 0.71$) improved to $\kappa = 0.79$ after calibration. In the main annotation phase, each evaluator independently assessed the remaining 450 questions from the stratified sample over four weeks, with weekly check-ins to address procedural questions. The annotation interface randomized question order and masked AI-generated explanations to prevent anchoring bias. For adjudication, questions with split decisions underwent synchronous discussion where evaluators reviewed evidence spans, consulted reference texts when needed, and reached consensus via majority vote (at least 3 of 5 agreeing). Final inter-rater reliability was Fleiss' $\kappa = 0.82$.

B Example Questions

B.1 Easy Difficulty Examples

Q1 (Easy): Khi trẻ mới tập ăn dặm, nên pha bột như thế nào?

(When a child is just starting solid foods, how should powder be mixed?)

A. Pha bột quá đặc (Mix powder too thick)

B. Pha bột loãng vừa phải ✓ (Mix powder moderately thin)

C. Pha bột quá loãng (Mix powder too thin)

D. Không pha bột (Don't mix powder)

Evidence: "Không nên pha bột quá đặc khi trẻ mới tập ăn dặm."

Explanation: Tests basic knowledge of traditional Vietnamese medicine regarding proper feeding practices for young children.

Q2 (Easy): Cách chữa đái dắt bằng lá cây nào?

(Which plant leaves are used to treat urinary tract infection?)

A. Lá chè xanh ✓ (Green tea leaves)

B. Lá đuối (Duối leaves)

C. Lá cà gai leo (Cà gai leo leaves)

D. Lá mò trắng (White mò leaves)

Evidence: "Lấy một nắm lá chè xanh rửa sạch, vẩy ráo nước sau đó mang giã nhỏ, rồi đổ nước đun sôi vào."

Explanation: Tests knowledge of traditional herbal remedies for common ailments.

B.2 Medium Difficulty Examples

Q3 (Medium): Cách điều trị chứng co giật do sốt ở trẻ là gì?

(What is the treatment method for fever-induced seizures in children?)

A. Tắm bằng nước có nhiệt độ cao (Bath with high temperature water)

B. Tắm bằng nước có nhiệt độ thấp hơn 2 độ so với thân nhiệt của trẻ ✓ (Bath with water 2 degrees lower than child's body temperature)

C. Cho trẻ uống thuốc kháng sinh (Give antibiotics)

D. Cho trẻ ăn nhiều đường (Feed child more sugar)

Evidence: "Tắm cho trẻ bằng nước có nhiệt độ thấp hơn 2 độ so với thân nhiệt của trẻ (có thể tắm nhiều lần như vậy)."

Explanation: Tests understanding of fever management techniques in traditional Vietnamese pediatric medicine.

Q4 (Medium): Cách chữa bệnh lang ben là gì?

(What is the treatment method for Tinea versicolor?)

A. Đắp kem Ciconten Plus lên vùng da bị bệnh (Apply Ciconten Plus cream)

B. Trộn củ riềng với rượu trắng nồng độ cao và đắp lên chỗ bị bệnh ✓ (Mix galangal root with high-proof alcohol and apply)

C. Ăn sống hoặc giã nhỏ tỏi pha với nước đun sôi để nguội uống (Eat raw or crush garlic with boiled water)

D. Uống 49 hạt đậu đen xanh lòng mỗi buổi sáng (Drink 49 black beans each morning)

Evidence: "Hãy lấy củ riềng rửa sạch, giã nhỏ và trộn với rượu trắng nồng độ cao (loại để ngâm thuốc) sao cho sền sệt, rồi đắp lên chỗ bị bệnh."

Explanation: Tests knowledge of traditional treatments for skin conditions using natural remedies.

B.3 Hard Difficulty Examples

Q5 (Hard): Cách điều trị bệnh bìu dái quá lớn ở trẻ em dưới 1 tuổi là gì?

(What is the treatment for excessively large scrotum in children under 1 year old?)

A. Đắp bột khô (Apply dry powder)

B. Ngâm bìu dái vào nước ấm và đốt lửa ngải ✓ (Soak in warm water and burn mugwort)

C. Trộn Phục long can với lòng trắng trứng (Mix Phục long can with egg white)

D. Giã nhừ 1 con giun đất lớn trộn với đường cát trắng (Crush earthworm with white sugar)

Evidence: "Trong vòng 1 tuổi, vào giờ Tý, ngày Đoan Dương (mùng 5 tháng 5) lấy thau đựng nước nóng đặt giữa nhà. Cho trẻ ngồi ngâm bìu dái vào nước ấm rồi bế trẻ đặt ngồi lên ngưỡng cửa, nước ở bìu dái sẽ in ngấn (vết) trên ngưỡng cửa. Đốt lửa ngải 3 lần trên vết nước ấy, lúc đốt hơi bìu dái trên khối ngải, bìu dái sẽ liền rút nhỏ lại ngay."

Explanation: Tests knowledge of complex traditional remedies with specific timing and procedural steps.

Q6 (Hard): Công dụng của bài thuốc trị cảm mạo, sưng phù, đau họng, chóng mặt, sốt, miệng lưỡi khô rát là gì?

(What is the effect of the remedy for cold, swelling, sore throat, dizziness, fever, and dry mouth?)

A. Thanh nhiệt, sơ phong, dưỡng âm (Clear heat, disperse wind, nourish yin)

B. Giải độc, tiêu sưng (Detoxify, reduce swelling)

C. Thanh lợi yết hầu, tiêu sưng, sinh nước bọt

giải khát ✓ (Clear throat, reduce swelling, generate saliva and quench thirst)

D. Trị bệnh cước chân mùa rét (Treat cold feet in winter)

Evidence: “Có tác dụng thanh lợi yết hầu, tiêu thũng, sinh nước bọt giải khát.”

Explanation: Tests understanding of complex therapeutic effects combining multiple medicinal properties.

C Error Analysis Details

Among 29 flagged questions (5.8% of the 500 validation sample), we identified four primary issue categories through systematic review. **Ambiguous distractor phrasing** (11 questions, 38%) involved non-standard terminology, regional variants, or overly similar options—for example, using both synonymous terms for hypertension as separate choices. **Outdated terminology** (8 questions, 29%) reflected archaic terms from older textbooks or inconsistent mixing of Sino-Vietnamese and pure Vietnamese terms, such as alternating between equivalent expressions for wind-cold syndrome. **Overly technical language** (6 questions, 21%) required specialized knowledge beyond typical practitioner training, including rare herb subspecies or obscure acupuncture point combinations. **Other issues** (4 questions, 12%) comprised factual errors, multiple defensible answers, and grammatical problems. These 29 flagged questions included 21 that underwent minor terminology standardization (counted in the 4.1% Minor Revision category) and 8 that were flagged for potential issues (1.7% Rejected/Flagged).

D Implementation Details

D.1 Text Chunking Configuration

We utilized the RecursiveCharacterTextSplitter (Chase, 2022) with the following parameters: chunk size = 2000 characters, overlap = 200 characters. This configuration balances context completeness with token limit constraints while ensuring smooth transitions between adjacent segments.

D.2 Teacher Model Sampling

Llama-3.1-70B-Instruct-AWQ generation parameters:

- Temperature: 0.6 (balances creativity with factual accuracy)
- Max tokens: 1024
- Stop sequences: [“”] (terminates at JSON closing delimiter)

- Top-p: 0.9

D.3 Student Model Sampling

Qwen2.5-32B-Instruct-AWQ validation parameters:

- Temperature: 0.2 (conservative sampling for stricter validation)
- Max tokens: 768
- Top-p: 0.85

D.4 JSON Output Schema

We enforce structured generation using the following schema with one-shot exemplar prompting:

```
{
  "question": str,
  "options": {"A": str, "B": str,
              "C": str, "D": str},
  "answer": str, // Must be A/B/C/D
  "explanation": str,
  "evidence": str // Quoted from context
}
```

References

- AI@Meta. 2024. [Llama 3 model card](#). *Meta AI Research*.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. [A large annotated corpus for learning natural language inference](#). In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 632–642, Lisbon, Portugal. Association for Computational Linguistics.
- Harrison Chase. 2022. LangChain: Building applications with LLMs through composability. <https://github.com/langchain-ai/langchain>. GitHub Repository.
- Yifan Chen, Ananya Pal, Hamid Palangi, and 1 others. 2024. [FreeMedQA: Benchmarking applied medical knowledge with free-response evaluations](#). *arXiv preprint arXiv:2405.09384*.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient fine-tuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115.
- Yanqiao Guo, Xiaoqing Chen, and 1 others. 2023. **BianCang: A traditional chinese medicine large language model**. *arXiv preprint arXiv:2310.15864*.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. **Distilling the knowledge in a neural network**. *arXiv preprint arXiv:1503.02531*.
- Junjie Hu, Sebastian Ruder, Aditya Siddhant, Graham Neubig, Orhan Firat, and Melvin Johnson. 2020. XTREME: A massively multilingual multi-task benchmark for evaluating cross-lingual generalization. In *Proceedings of the 37th International Conference on Machine Learning*, pages 4411–4421. PMLR.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, and 1 others. 2023. **Mistral 7B**. *arXiv preprint arXiv:2310.06825*.
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hsuan Weng, Hanyi Fang, and Peter Szolovits. 2021. **What disease does this patient have? a large-scale dataset with symptom-to-diagnosis and treatment rationales**. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3404–3417, Online. Association for Computational Linguistics.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. 2019. **PubMedQA: A dataset for biomedical research question answering**. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, Hong Kong, China. Association for Computational Linguistics.
- Alistair Johnson, Tom J. Pollard, and Roger G. Mark. 2023. Biomedical text mining and natural language processing in low-resource languages. *Journal of Biomedical Informatics*, 138:104280.
- J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, pages 9459–9474. Curran Associates Inc.
- Ziyang Li, Mengzhou Wang, and 1 others. 2024. **TCM-Ladder: A benchmark for evaluating large language models on traditional chinese medicine**. *arXiv preprint arXiv:2402.08159*.
- Chien Van Nguyen and Liem Tan Nguyen. 2024. **Vistral: Vietnamese Mistral 7B**. <https://huggingface.co/Viet-Mistral/Vistral-7B-Chat>. Hugging Face Model Hub.
- Van Thanh Nguyen and Minh Hoang Le. 2018. Traditional medicine knowledge representation and reasoning: A survey. In *Proceedings of the 10th Asian Conference on Intelligent Information and Database Systems*, pages 365–375. Springer.
- Quoc Long Pham and Thi Hong Nguyen. 2020. Vietnamese traditional medicine: Historical development and contemporary practice. *Journal of Ethnopharmacology*, 252:112589.
- Qwen Team. 2024. **Qwen2.5 technical report**. *arXiv preprint arXiv:2409.12191*.
- Atnafu Lambebo Tonja, Vijeta Mullachery, and 1 others. 2024. **AfriMed-QA: A pan-african multi-specialty medical question answering benchmark**. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.
- Quan Tran, Long Phan, Tuan Vo, and Hieu Nguyen. 2023. **VinaLLaMA: LLaMA-based Vietnamese large language models**. *arXiv preprint arXiv:2312.11011*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837.
- Qian Zhang, Junnan Chen, and 1 others. 2024a. **MMed-Bench: A multilingual medical benchmark for large language models**. *arXiv preprint arXiv:2402.09856*.
- Yucheng Zhang, Zhe Liu, and 1 others. 2024b. **MedR-GAG: Retrieval-augmented generation with adaptive selection for medical question answering**. *arXiv preprint arXiv:2403.12807*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623.